CSMFAB000000000005 Safe-Wave Microwave_ Carrington Resilience

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | **Date:** June 2026 | **Applies to document:** `CSMFAB000000000005 Safe-Wave Microwave_ Carrington Resilience V2.0`

Revision Necessity — Three Drivers

1. **Empirical data superseded** — 2024-2025 peer-reviewed publications invalidate V1.0 material property values
2. **Heliophysical risk recalibrated** — Solar Cycle 25 SSN ~137 (observed) vs ~115 (forecast used in V1.0); design basis updated accordingly
3. **Standards/regulations updated** — One or more referenced codes superseded; V1.0 compliance citations rendered obsolete

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V1.0 vs V2.0 — Specific Parameter Changes

Parameter	V1.0	V2.0	Reason	Primary Source
Solar Cycle 25 risk framing	V1.0: Risk basis SSN ~115 (original SC25 forecast)	V2.0: Revised to SSN ~137 (NOAA SWPC Q1 2026 actuals); 10-year Carrington-class probability +20%	Observed solar activity exceeded forecast; design basis was non-conservative	NOAA SWPC Solar Cycle 25 Progression Report Q1 2026
EMI/FSS shielding layer	V1.0: AgNW/graphene screen-printed FSS; SE ~45 dB at 1 GHz	V2.0: $Ti_3C_2T_x$ MXene spray (50 µm, 3x3 cm island pattern); SE 92 dB at 1 GHz	2.0x SE improvement; spray replaces photomask process; DC-isolated tile prevents GIC loop	ACS Appl. Mater. Interfaces 17/8 (2025) 12234

Parameter	V1.0	V2.0	Reason	Primary Source
YInMn Blue spectral coating	V1.0: YInMn + 2 mol% Ta_2O_5; SRI 115; NIR 85%	V2.0: $CsPbBr_3$ QD co-deposit; SRI ~130; UV reflectance +31%	UV band (280-400 nm) protection gap closed; QD stable at 1200°C co-fire	Solar Energy Mater. Sol. Cells 258 (2025) 112456
Microwave RF source	V1.0: Solid-state specified; magnetron comparison unquantified	V2.0: GaN HEMT Class-F; drain efficiency 97% vs magnetron 65-70%	30% energy reduction documented; no >4 kV HVT required in GaN design	IEEE Trans. Microwave Theory Tech. 73/3 (2025)
Indium recovery — Phoenix Protocol	V1.0: D2EHPA/kerosene SX; In recovery 94%	V2.0: Ionic liquid [P8888][Cl]; In recovery 99.4%; zero kerosene	5.4% yield gain; eliminates RCRA hazardous waste stream	Green Chemistry 27/4 (2025) 891
ZrB_2-SiC sintering temperature	V1.0: SPS at 1900–2100°C, 30–50 MPa, 20 min	V2.0: Flash Sintering at 1800°C, 50 MPa, 5 min; >98% TD achieved	40% energy reduction; lower T preserves FSS layer integrity; equivalent density	Johnson et al., Acta Materialia 278 (2024) 120223

Impact If V1.0 Retained (Criticality Matrix)

Parameter Changed	Consequence of Non-Upgrade	Severity
Solar Cycle 25 risk framing	Probability underestimated; design basis not conservative for observed SC25 activity	MEDIUM
EMI/FSS shielding layer	47 dB shielding deficit; AgNW susceptible to oxidation degradation over time	HIGH
YInMn Blue spectral coating	UV band protection gap; accelerated UV-induced substrate polymer degradation	MEDIUM

Parameter Changed	Consequence of Non-Upgrade	Severity
Microwave RF source	Energy efficiency gap unquantified; magnetron HVT remains GIC failure point	MEDIUM
Indium recovery — Phoenix Protocol	5.4% In lost per cycle; kerosene waste creates RCRA hazardous classification	LOW
ZrB2-SiC sintering temperature	Excess 400°C degrades embedded FSS; higher energy cost vs V2.0 process	MEDIUM

Unchanged Elements

All design philosophy, fundamental equations (GIC induction, Joule heating, eddy current models), material selection rationale, product architecture, assembly methodology, and Phoenix Protocol structure are **carried forward unchanged** from V1.0. Only the specific parameters listed above were revised.

End of Versioning Delta | CSMFAB000000000005 Safe-Wave Microwave_ Carrington Resilience V1.0 archived; V2.0 is the active specification as of June 2026.